TRIANGULATION

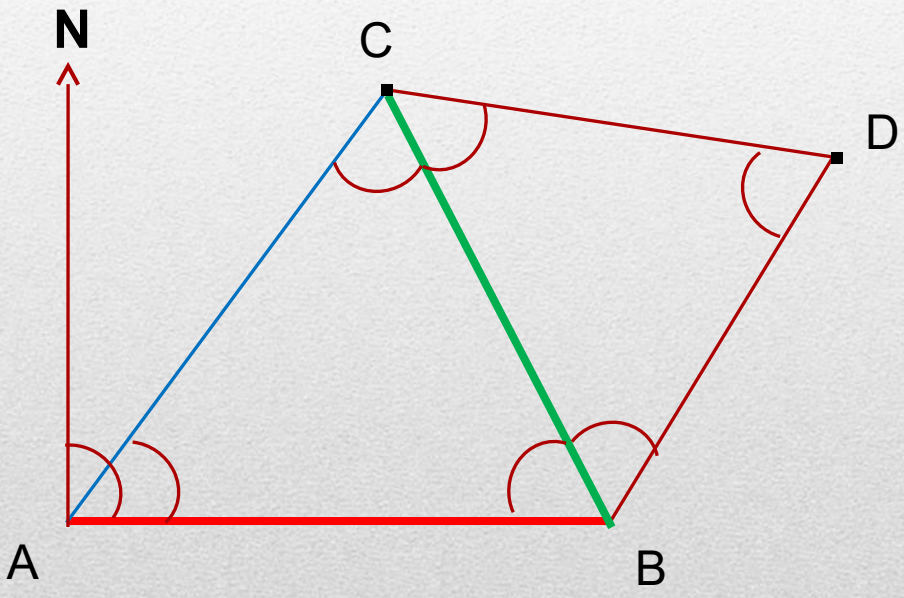
ENGR AMEER HAMZA

TRIANGULATION

- It is one of the methods of providing control in an area which is to be surveyed
- It is based on the Trigonometric proposition that if one side and three angles of a Triangle be known, the remaining sides can be computed by the application of Sin Rule i.e.

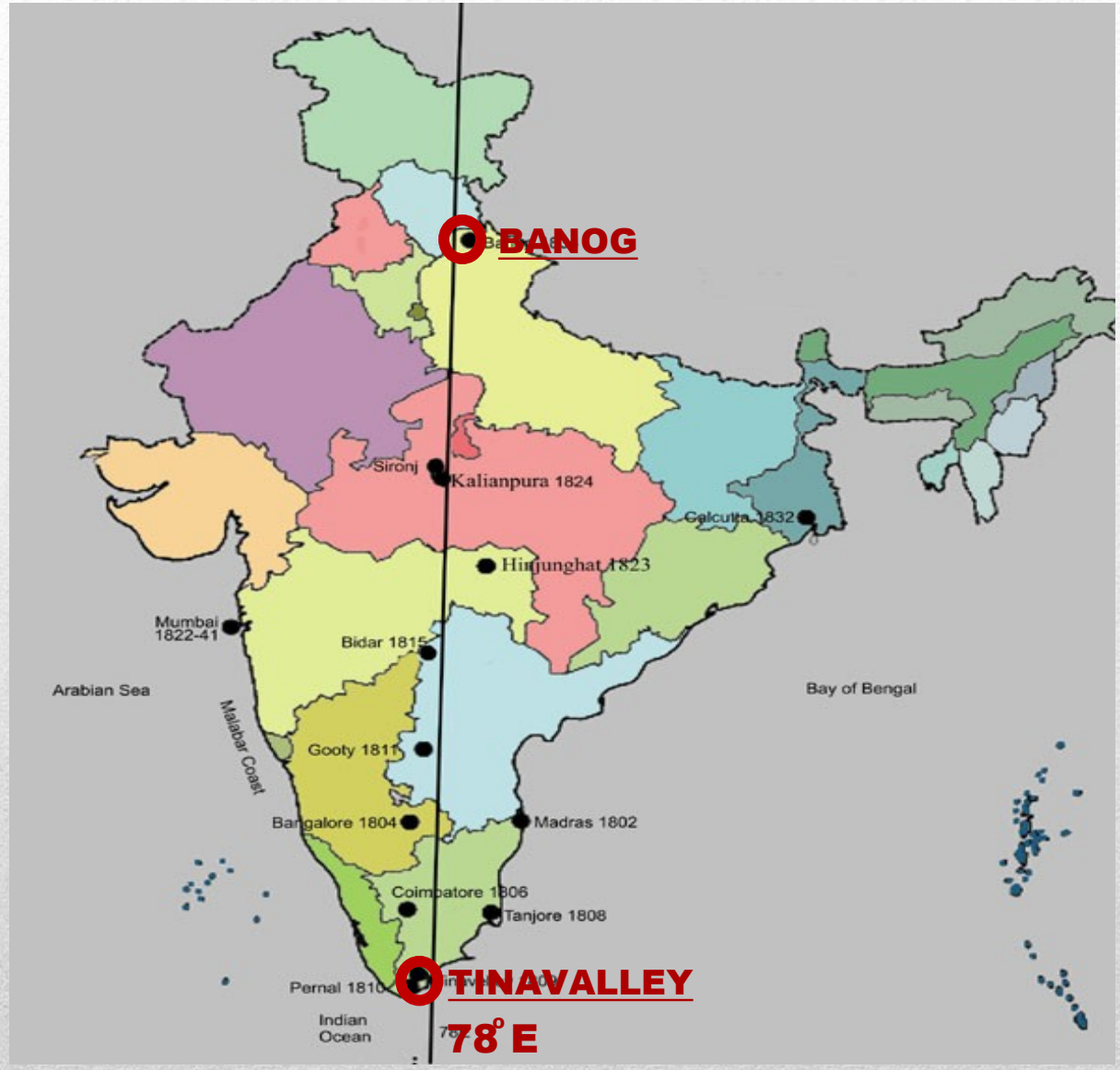
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

TRIANGULATION



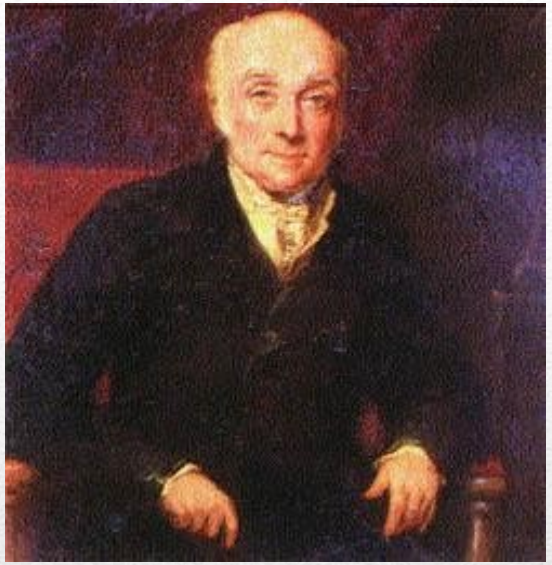
BACKGROUND

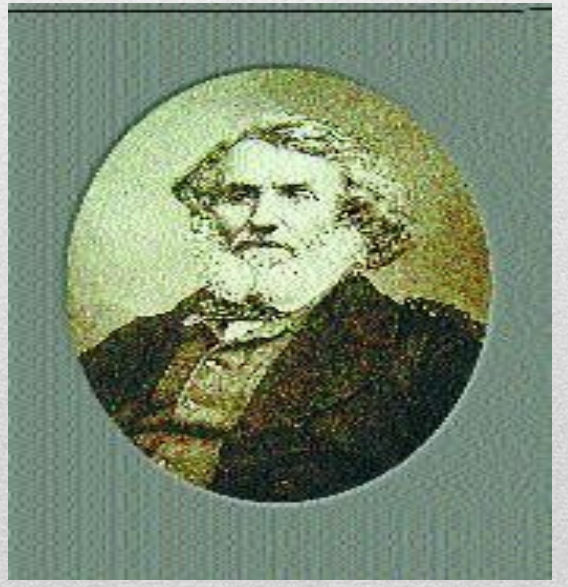
- Triangulation has been the classical method of providing control till 1960s throughout the World
- In the Indian Subcontinent it has been used for monumental applications by the British surveyors for:
  a. Measurement of a Great Arc for determining the shape of Earth



The Great Trigonometric Survey was an attempt to measure an arc of 78° E from Tinavelley to Banog.

1. Started by Lambton in 1802
2. Everest joined him in 1818
3. On 20 Jan 1823 Lambton died and Everest took over GTS
4. Everest completed work in 1866
5. Establishment of control by iron grid method





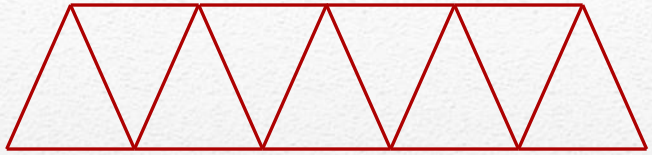
THE SEQUENCE OF DIFFERENT STEPS IN THE PROCESS OF TRIANGULATION

1. Selection of suitable points as triangulation stations in the area to be surveyed
2. Measurement of a Base Line
3. Measurement of all the angles of the triangle and adjustment of misclosures.
4. Computations of lengths of other two sides of the triangle

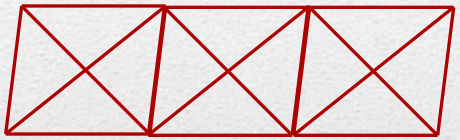
5. Computation of coordinates of unknown point
6. Taking one computed side as Base for the next triangle and measurement of all the angles and so on
7. Measurement of a check Base at an interval of 200 to 500 Km
8. Determining Azimuth at the intervals of 250 Km by astronomical observations. Determining of Longitude and latitude are also necessitated at times

TRIANGULATION FIGURES

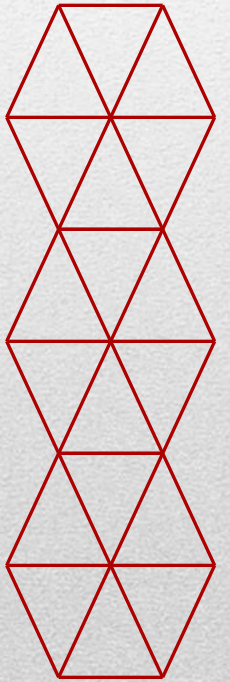
1. Triangles



2. Quadrilaterals



3. Pentagon or Hexagons
   (Polygons)



Chain of braced quadrilaterals is the best followed by centered point polygon

CLASSIFICATION

Primary

a. Initial survey of an area
b. First order standard accuracy
c. Side length 50—100km or as long as possible depending upon terrain and atmosphere
d. Triangle misclosure 1”—2”

CLASSIFICATION

Secondary

a. Breakdown of primary triangle
b. Second order accuracy
c. Side Length 20—50Km
d. Triangle misclosure 4”—5”

CLASSIFICATION

Tertiary

a. Further break down of primary and secondary work to provide control for Topographic survey
b. Third order accuracy
c. Side length 5—20Km
d. Triangle misclosure 5”—10”

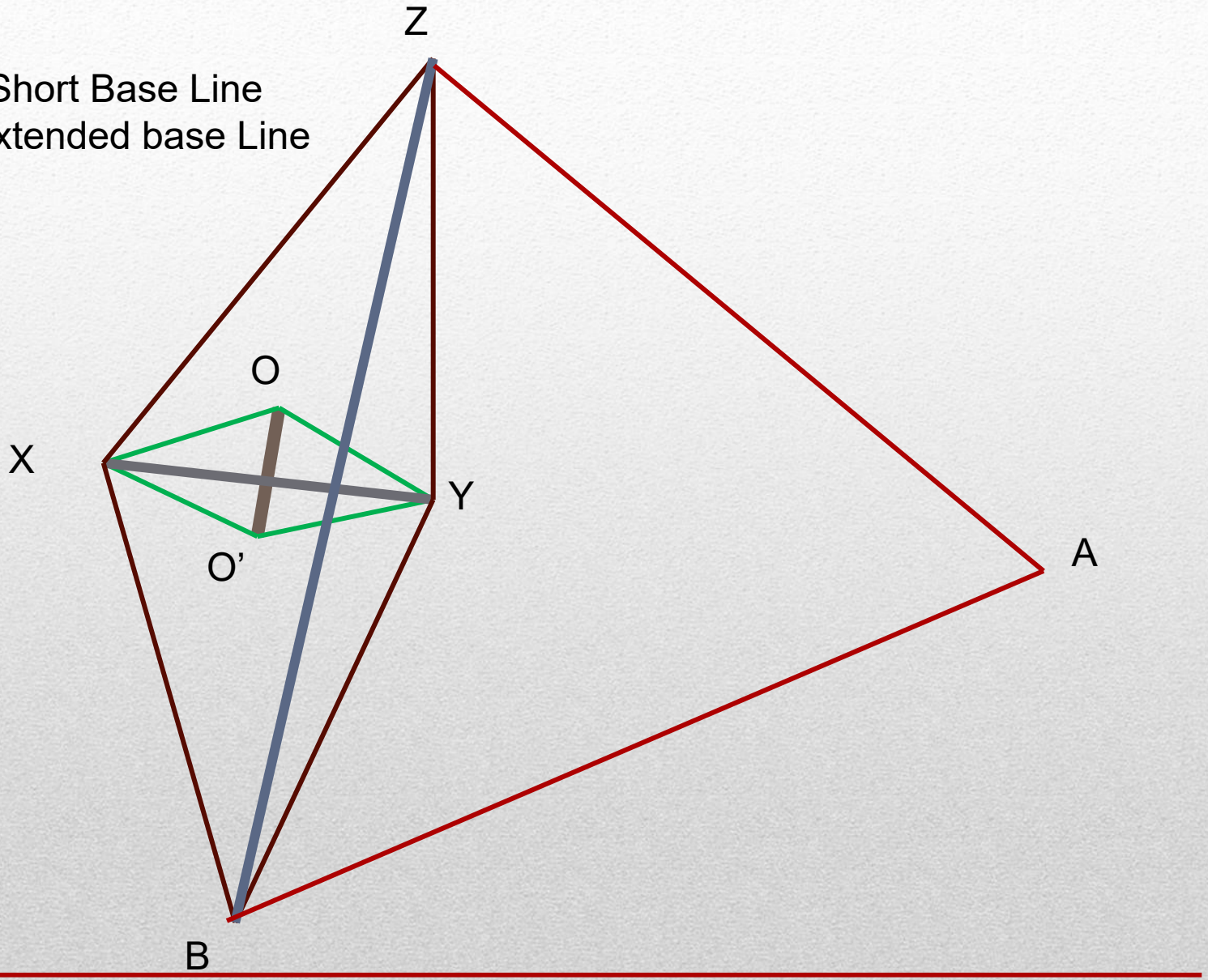
ABSOLUTE POSITION, ORIENTATION & SCALE

- Accurate measurement of Base will set the scale
- Determining the Azimuth will fix the orientation
- Determining the Longitude and Latitude of one of the base terminals will fix the position on the earth's surface

BASE & BASE EXTENSIONS

- Conventionally the Base Lines were measured with Invar Tapes and it was not possible to measure very long distances with them
- Therefore shorter Bases were measured and then these were extend by a series of well conditioned figures

OO’ Short Base Line
BZ Extended base Line



PRESERVATION OF ACCURACY

Accuracy is preserved by

a. Check Bases at intervals of 200 to 500Km
b. Azimuth is checked at intervals of 250Km
c. Using well conditioned figures. Angle (30° —120°) but an equilateral triangle is ideal and square equilateral is ideal
d. Fully observing all closed figures

DESIGN OF TRIANGULATION

a. Series of connecting chains or Grid iron system. (Australia, India, USA)
b. A continuous mesh of triangles or the central system (UK)



Fig. Map of the Indian subcontinent showing the triangulations and survey carried out by the Survey of India in the 19th century. The Great Arc passes through the "centre" of India. Also shown are secondary survey projects and measurements of the Himalayan peaks

Steps of Triangulations

RECONNAISSANCE

1. Examination of country to be surveyed
2. Suitable site for Base Line
3. Positions for triangulation stations
4. Determining
    a. Inter-visibility
    b. Clearance required in line of sight for inter-visibility

RECONNAISSANCE

5. Information regarding
   a. Access to station
   b. Transport facilities
   c. Supply of food, water etc
   d. Camping ground or suitable accommodation

SELECTION OF STATIONS

1. Inter-visible
2. Should form well conditioned triangles
3. Should be easily accessible
4. Useful for detail survey
5. Convenient/suitable length of line of sight

STATION MARKS

- Should be permanently marked with copper or bronze tablet on which its name and year be stamped
- Should be set on concrete monument or stable rock
- In earth an additional mark be placed about 30 inches below surface in stone or concrete

TOWERS

- In flat areas or under peculiar conditions towers may be necessary
- Calculation of height of tower needed for clearance of line of sight
- It increases logistic problems

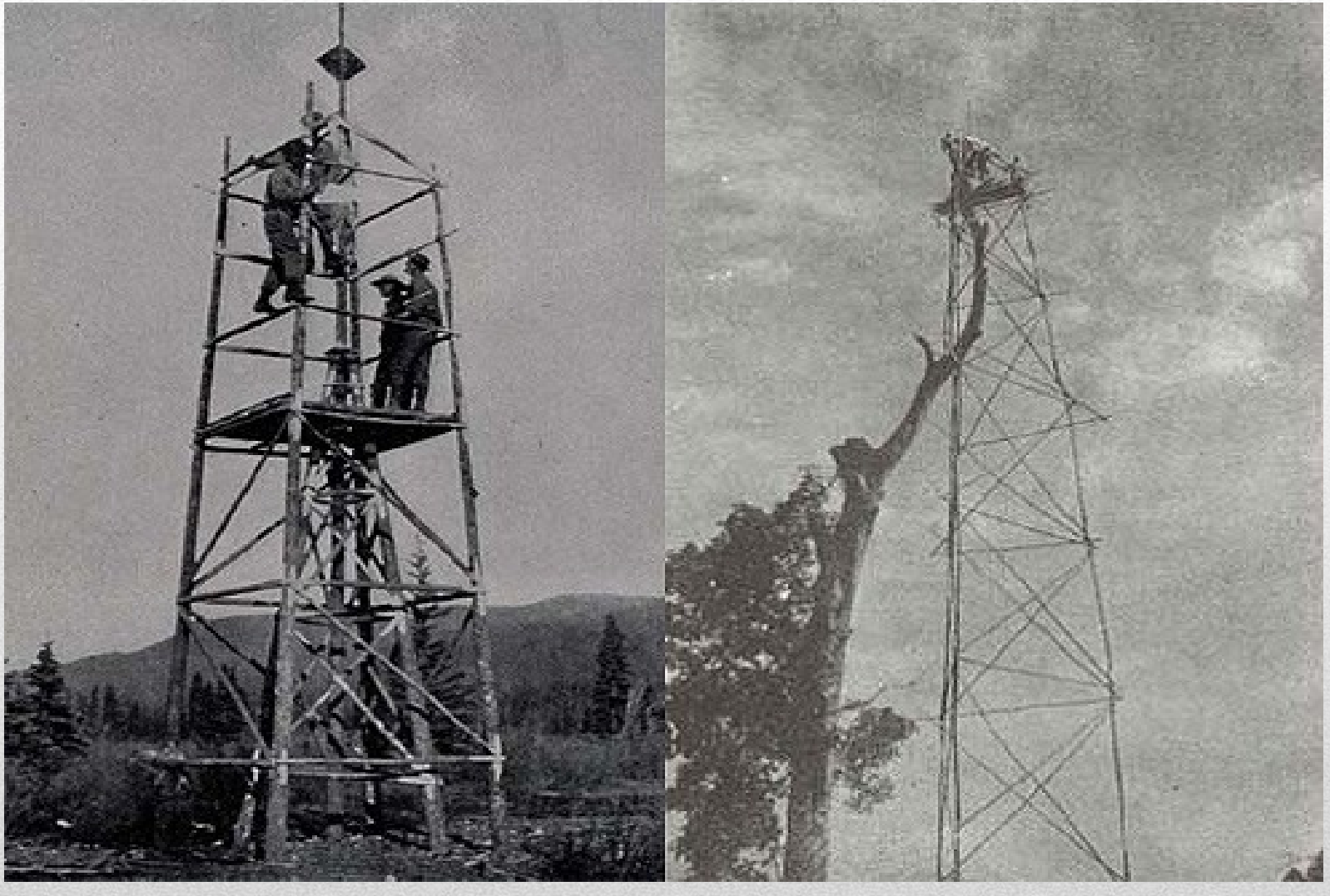


One of the Great Arc's towers, minus its topmost railings, at Begarazpur to north of Delhi





BILBY Tower



TIMBER Tower

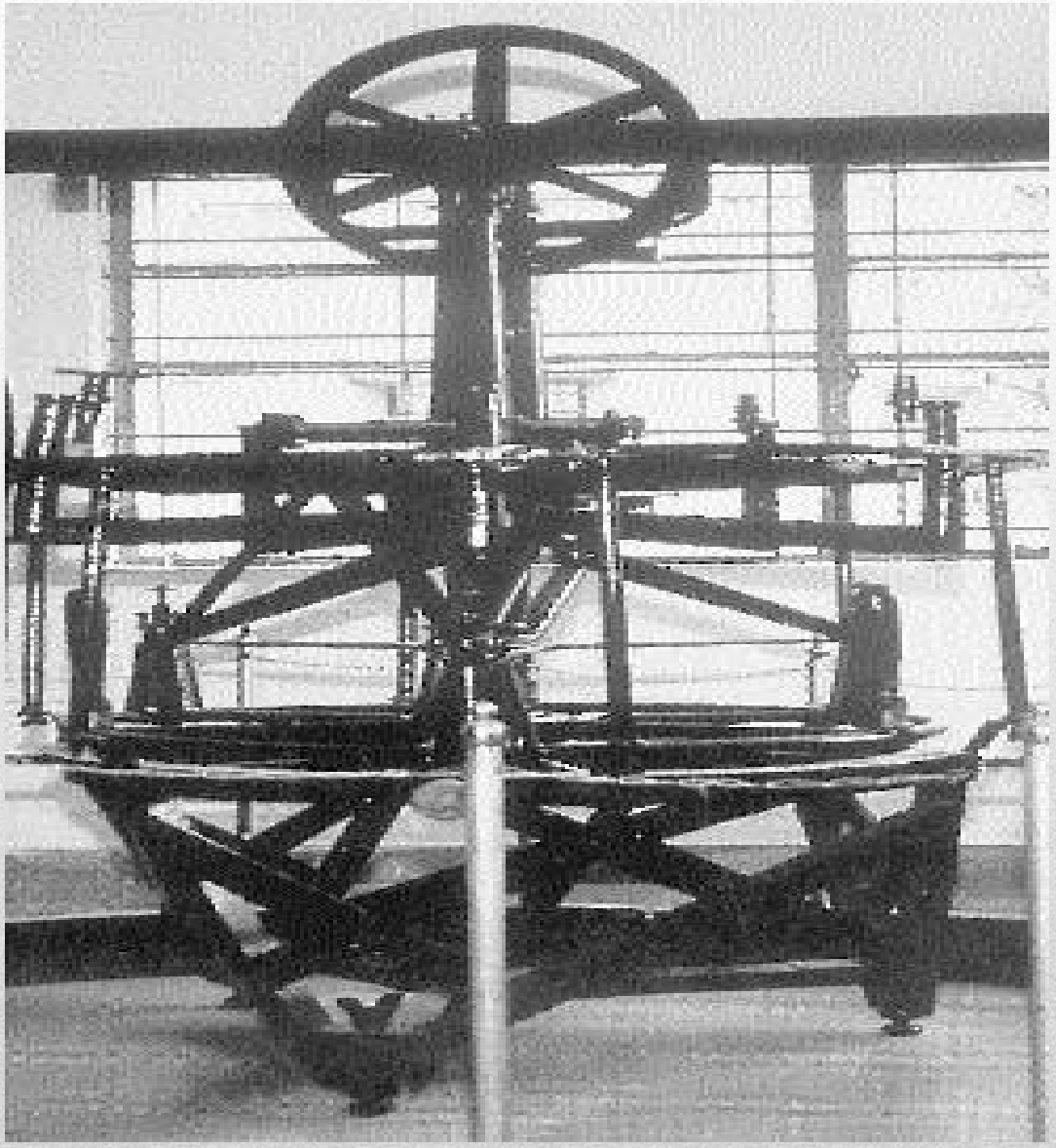


PIPE Tower

COMPUTATION & ADJUSTMENTS

Observations are adjusted for

a. Removing misclosures
b. Consistency



Lambton's Great Theodolite weighed half a tonne. Severely damaged on various occasions, it was rebuilt and is now in the Survey of India's headquarters in Dehra Dun.

